Facial Soft Tissue Repositioning With Neuromodulators: Lessons Learned From Facial Biomechanics

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Abstract

Background: Neuromodulators have proven efficacy in reducing facial rhytides and have also been reported to improve jawline contour and the appearance of platysmal bands. Lifting effects of the tail of the eyebrow are expected outcomes when targeting the lateral periorbital region, underscoring the versatility of neuromodulator treatments.

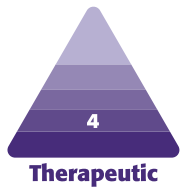
Objectives: The aim of this study was to analyze the clinical effectiveness of a novel neuromodulator-based injection algorithm with regards to its ability to reposition the middle and lower facial soft tissues.

Methods: Seventy-five study participants (8 males, 67 females) with a mean [standard deviation] age of 37.5 [8.5] years were injected with neuromodulators in the subdermal plane of the mandibular soft tissues following a standardized algorithm. Live rating of clinical appearance was performed, as well as volume change and skin vector displacement measured by 3-dimensional imaging at baseline, Day 14, and Day 30.

Results: Three-dimensional volume analysis revealed an increase in midfacial volume by 0.46 mL, and a decrease of the lower facial volume by 0.30 mL compared with baseline. Additionally, an improvement of midfacial fullness (by 0.13) and jawline contour (by 0.44) was reported on clinical rating scales at Day 30 compared with baseline.

Conclusions: Facial soft tissues can be repositioned during the 30-day follow-up period following a neuromodulator treatment; this was reflected through an increase in midfacial volume as well as through a decrease in lower facial volume. The novel injection algorithm presented can provide a safe and effective option for patients desiring improvement of midfacial fullness and jawline contour with neuromodulator treatment alone.

Level of Evidence: 4



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Neuromodulator injections are a widely accepted minimally invasive aesthetic treatment for the face. According to the 2020 annual statistics report released by The Aesthetic Society, more than 2.6 million procedures were performed, doubling the use of dermal fillers for that year.[1] Neuromodulators are primarily utilized for the treatment of facial rhytides by temporarily reducing the contractility of the targeted muscle. Recent research has revealed that by incorporating anatomic knowledge into neuromodulator treatment algorithms, the rate of adverse events can be reduced and effectiveness increased. Examples of this are the concept of the line of convergence,[2] the description of the subfrontalis fascia,[3,4] and deep forehead injections for the treatment of horizontal forehead lines.[5]

However, in daily clinical practice the widespread use of neuromodulators in the upper face may lower the position of the eyebrow by targeting its elevator (frontalis muscle) or elevate the tail of the eyebrow by targeting its depressor (lateral orbicularis oculi muscle).[6,7] Altering the eyebrow position with neuromodulators is possible due to a profound understanding of the underlying anatomy, the equilibrium between brow elevator and depressors, and the subsequent clinical effects.

In 2007, the "Nefertiti lift" was first reported in the literature by Levy, who described a neuromodulator injection technique in a case series of 130 patients where the platysma muscle was targeted "along and under each mandible and to the upper part of the posterior platysmal band" to enhance jawline and neck contour.[8] Potentially due to the occurrence of 2 cases of dysphagia in the original study, this injection technique was modified[9-11] but its alterations still aimed to improve the appearance of the transition between lower face and neck by increasing jawline contour and by reducing platysmal banding.

Novel anatomic research has introduced the term "facial biomechanics" which is based on the collective interplay of facial structures during various facial expressions and aging.[12] One of those biomechanical concepts includes the fascial connection between the platysma, the superficial musculoaponeurotic system (SMAS), the orbicularis oculi muscle, and the zygomaticus major muscle.[13-17] The latter has been shown to be connected with the transverse facial septum, a structure that forms the inferior boundary of the midfacial fat compartments.[18]

The present study sought to evaluate the effects of reducing the caudal pull of the platysma, the major facial depressor, on the volume of the middle face (hypothesis 1), whether the lower and midfacial soft tissue can be repositioned (hypothesis 2), and if the contour of the jawline can therefore be enhanced (hypothesis 3). To explore these hypotheses, this clinical, interventional prospective study was designed and conducted utilizing neuromodulators in a novel injection algorithm.

METHODS

Study Design

This study was designed as a multicenter, interventional, prospective clinical investigation that aimed to identify the potential effects (if any) of a new injection algorithm utilizing neuromodulators on midfacial volume and jawline contour. The protocol was designed centrally and distributed to the 3 participating study centers, which were bound to the protocol for the injection technique only but were free to choose the amount or type of neuromodulator administered. The study centers in return were not involved in data analysis or data interpretation; this specific study design is thought to increase objectivity and to decrease investigator influence on study outcome. The study was conducted between July 2020 and July 2021 and received IRB approval from the Medical Faculty, University of Belgrade, Serbia (number: 9P21J42AN0319/2020).

Study Centers and Patient Selection

The participating study centers were based in Colombia (CH Dermatologia, Medellín), Brazil (Cirurgia Plastica, Belo Horizonte), and Serbia (Rea Medika, Belgrade). Patients included in this study had to be younger than 50 years old with a BMI of less than 30 kg/m^2. Exclusion criteria included history of facial surgeries, neuromodulator treatment of the face and/or neck within 6 months of enrollment, previous facial soft tissue filler injections, or any other type of facial aesthetic procedures that could influence normal facial anatomy or physiologic facial mobility. Informed consent was obtained from all study participants prior to inclusion for the use of their images and personal data for study purposes.

Injection Algorithm

The novel "toxin lift" injection algorithm was designed to increase the midface volume and to enhance the contour of the jawline, without directly influencing the neck or the presence of platysmal bands. Per facial side, 4 injections were performed, each located 1 cm above the palpable superior margin of the bony mandible starting at the oral commissure and extending equidistantly to the angle of the mandible (Figure 1). The neuromodulator injections were applied subdermally with the tip of the needle pointing laterally (toward the angle of the mandible) to avoid the depressor labii inferioris muscle during the first (ie, the most medial) injection. The amount of neuromodulator injected was 2.0 or 2.5 international units (IU), resulting in a total of 8 or 10 IU per facial side and in 16 or 20 IU for the total face. The choice whether to administer 2 or 2.5 IU was based on the clinical judgment of the injector. The neuromodulators administered in this study were abobotulinumtoxinA

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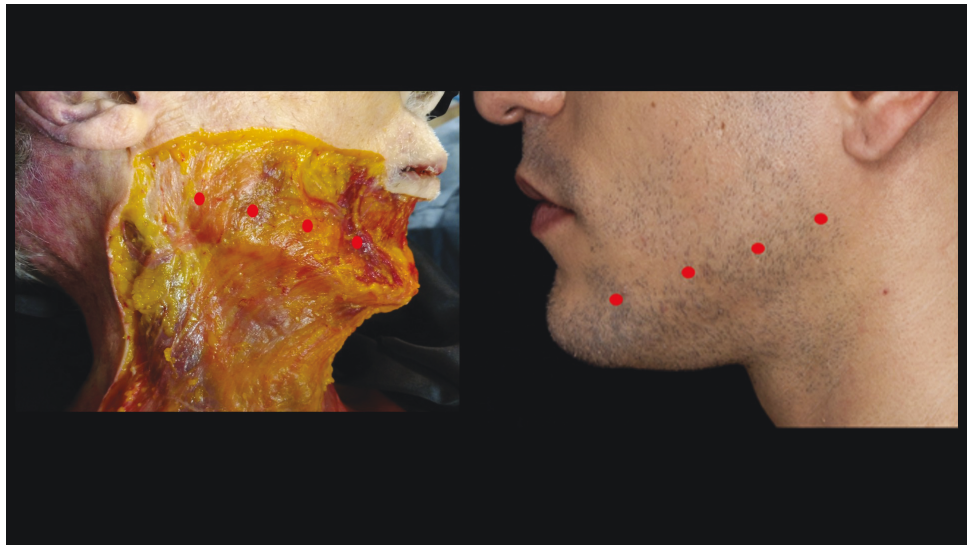

Figure 1. Standardized injection algorithm shown on a 30-year-old male patient and exemplified on a nonembalmed cadaveric dissection. On each facial side, a total of 4 points, each located 1 cm cranial to the palpable superior margin of the bony mandible, were injected with neuromodulators. The injections were administered at equal distances between the oral commissure and the mandibular angle.

(Dysport; Galderma, Lausanne, Switzerland) in a 20 IU per 0.1 mL dilution, OnabotulinumtoxinA (Botox; Allergan, Irvine, CA) in a 5 IU per 0.1 mL dilution, and IncobotulinumtoxinA (Xeomin; Merz Pharmaceuticals, Frankfurt, Germany) in a 5 IU per 0.1 mL dilution. All products utilized in this study were provided by the participating study centers without financial support from industrial partners. For the included patients, however, the neuromodulator treatment was free of charge.

OUTCOME ASSESSMENT

Semiquantitative Outcome Scoring

Semiquantitative outcome assessment of the newly developed neuromodulator injection algorithm relied on validated midfacial volume and jawline contouring scales. Both scales ranged from 0 to 4, best to worst, with both representing: 0 = none, 1 = mild, 2 = moderate, 3 = severe, 4 = very severe.[19,20] The scales were assessed in person by the treating physician and by an independent observer with an aesthetic background (colleague or first assistant to the injector).

A self-designed scale for gummy-smile severity was also utilized by the treating physician and by the same independent observer. The scale scores ranged from 0 to 4, best to worst, with: 0 = none, 1 = mild, 2 = moderate, 3 = severe, 4 = very severe.

Objective Outcome Scoring

Objective outcome assessment relied on 3-dimensional (3D) imaging of study participants in repose and with a Duchenne-type smiling facial position, as previously described.[21] In brief, standardized full-face 3D images were obtained with a Vectra H2 camera system (Canfield, Parsippany, NJ). The images were automatically aligned to the baseline image and differences in volume (volume change in mL) and differences in skin position (skin displacement in mm) were calculated following the internal computational algorithm of the Mirror software toolkit (Canfield) for the following facial regions: medial face, lateral face, lower face, and middle face (Figure 2).

Statistical Analysis

The outcomes assessed in this study were the semiquantitative midfacial volume and jawline contouring scores and the objectively measured change in volume and skin displacement of the investigated facial regions. Comparisons were made between baseline (ie, before neuromodulator injection) and the assessments performed at Day 14 and at Day 30 following the treatment (Figures 3, 4).

Assessment scores obtained by the treating physician and by the independent observer were averaged and the mean value was used for statistical analyses.

Pairwise (within-person) comparisons were calculated for differences between time points (baseline, Day 14, Day 30) by paired *t*-test. Age was dichotomously stratified into younger and older study participants related to the average age of the entire study sample (37.5 years): n = 36, age ≤37 years and n = 39, age ≥38 years for age-related comparisons. All calculations were run with SPSS Statistics 23 (IBM, Armonk, NY) and differences were considered statistically significant at a probability level of ≤0.05 to guide

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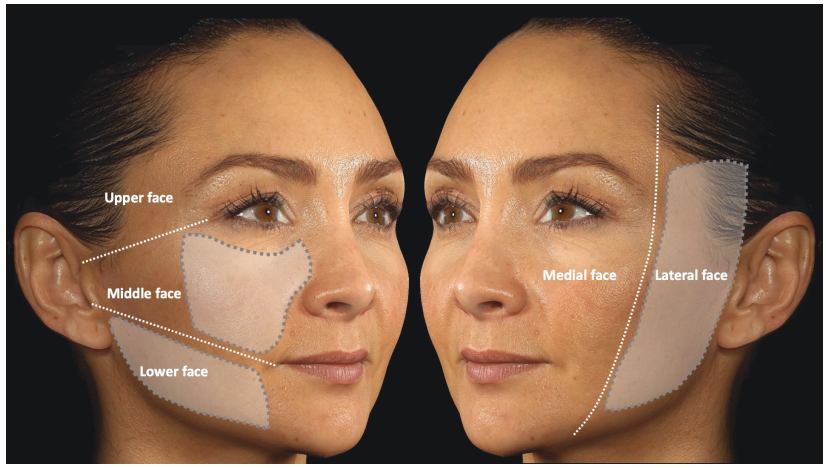

Figure 2. Three-dimensional scan photographs of a 42-year-old female study participant showing the facial regions in which volumetric and skin vector displacement analyses were performed: middle face, lower face, medial face, and lateral face.

Figure 3. Comparison of two 3-dimensional scan photographs of a 39-year-old female study participant in repose between baseline and Day 30. Note the change in midfacial volume.

Figure 4. Comparison of two 3-dimensional scan photographs of a 43-year-old female study participant in smiling facial position between baseline and Day 30. Note the change in midfacial volume and the exposure of the upper teeth during smiling.

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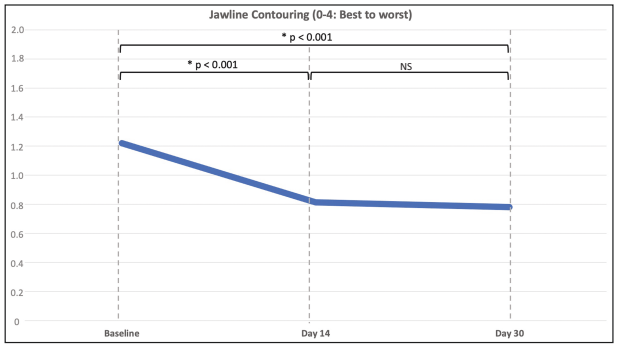

Figure 5. Line graph showing the mean scores of the jawline contouring scale during the 30- day follow-up period. Significance levels are provided for reference; NS, not statistically significant.

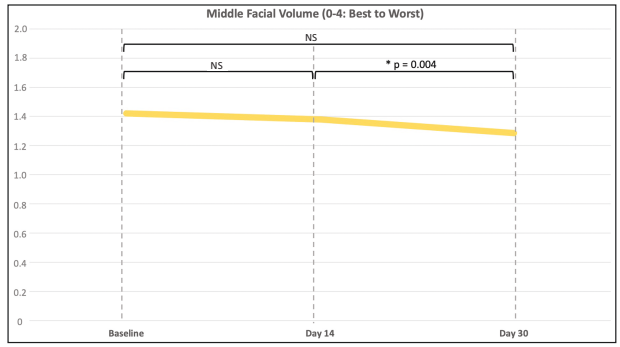

Figure 6. Line graph showing the mean scores of the midfacial volume scale during the 30-day follow-up period. Significance levels are provided for reference; NS, not statistically significant.

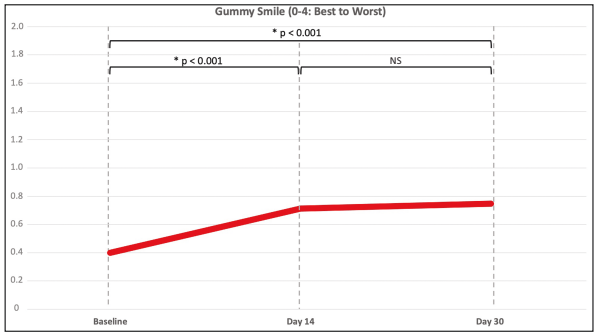

Figure 7. Line graph showing the mean scores of the gummy-smile severity rating during the 30-day follow-up period. Significance levels are provided for reference; NS, not statistically significant.

conclusions. Results are presented as mean [standard deviation] values.

RESULTS

Demographic Description

A total of 75 participants were included in this study, of whom 67 were females (89.3%). The mean age of the total study sample was 37.5 [8.5] years (range, 20-50 years) and the mean BMI was 22.90 [2.7] kg/m2 (range, 17.3-29.7 kg/m^2). Participants were classified according to Fitzpatrick skin type: Type II, n = 35 (46.7%); Type III, n = 29 (38.7%); Type IV, n = 7 (9.3%); and Type V, n = 4 (5.3%). All study participants initially included in this study (n = 75) completed the study protocol consisting of the 3D imaging-based measurements and the clinical evaluations at baseline, Day 14, and Day 30.

The mean scores prior to the neuromodulator treatment (ie, baseline scores; 0-4, best to worst) were 1.22 [0.9] (range, 0-4) for jawline contouring, 1.42 [0.8] (range, 0-3.5) for midfacial volume, and 0.4 [0.7] (range, 0-3) for gummy-smile severity without any statistically significant difference between genders ($P > 0.05$) for all parameters. Baseline severity for jawline contouring and for midfacial volume scores varied statistically significantly between the administered neuromodulator types ($P = 0.001$ and $P = 0.035$, respectively). At baseline, there was a statistically significant difference between younger (≤37 years; n = 36) and older (≥38 years; n = 39) study participants with regard to jawline contouring (0.74 [0.6] vs 1.67 [0.8], $P < 0.001$) and midfacial volume (0.92 [0.5] vs 1.88 [0.8], $P < 0.001$).

INJECTION PROCEDURE AND ADVERSE EVENTS

The average amount of neuromodulator used per total face was 16.43 [1.24] IU (range, 16.0-20.0 IU) without a statistically significant difference between genders ($P = 0.861$). IncobotulinumtoxinA was used in n = 38 patients (50.7%), AbobotulinumtoxinA was used in = 20 patients (26.7%), and OnabotulinumtoxinA was used in n = 17 patients (22.7%). During the 30-day follow-up period no (n = 0; 0%) adverse events (dysphagia, asymmetric smile, reduced oral competence, hematoma, pain) were observed. Gummy smile severity worsened significantly at Day 30 to 0.75 [1.2] (range, 0-4) ($P < 0.001$).

SEMIQUANTITATIVE OUTCOME SCORES

Thirty days after the initial neuromodulator treatment, midfacial volume increased to 1.30 [0.98] (range, 0-4) but failed to reach statistical significance ($P = 0.096$), whereas jawline contouring improved statistically significantly to 0.78 [0.8] (range, 0-3) ($P < 0.001$) (Figures 5-7). The change between baseline and Day 30 for younger vs older study participants was 0.19 [0.7] vs 0.08 [0.7] ($P = 0.461$) for their midfacial volume and 0.28 [0.5] vs 0.59 [0.5] ($P = 0.01$) for

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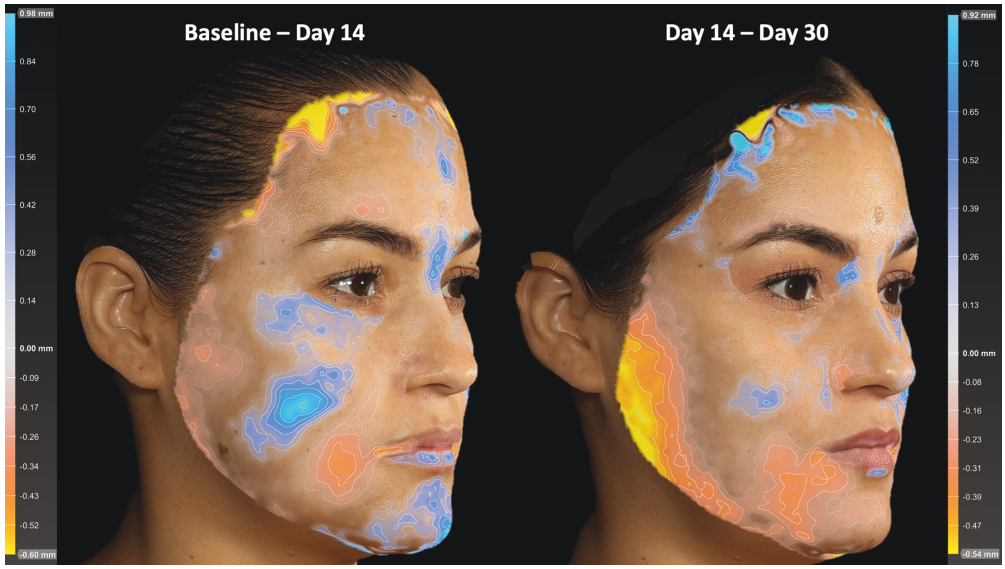

Figure 8. Three-dimensional scan photographs of a 34-year-old female study participant showing the volumetric changes between baseline and Day 14, and between Day 14 and Day 30, respectively. Blue represents volume increase and orange represents volume decrease as indicated by the scale on the sides.

jawline contouring. This indicates a greater improvement for older study participants in their jawline contour following the treatment but no age-related difference in the improvement for their midfacial volume.

3D IMAGE ANALYSES

Volume Changes

Comparing the 3D imaging values between baseline and Day 30, the volume of the middle face statistically significantly increased by 0.46 [0.9] mL (range, –1.68 to 2.80 mL) ($P < 0.001$) and the volume of the lower face decreased by 0.30 [1.5] mL (range, –6.25 to 3.16 mL) but failed to reach statistical significance ($P = 0.115$). In the medial face (ie, medial to the line of ligaments), the volume statistically significantly increased between baseline and Day 30 by 0.34 [1.2] mL (range, –4.80 to 2.40 mL) ($P = 0.025$) whereas in the lateral face (ie, lateral to the line of ligaments) the volume decreased by –0.07 [1.1] mL (range, –3.21 to 2.63 mL) but failed to reach statistical significance ($P = 0.596$) (Figure 8). No statistically significant differences in the measured volume changes were observed between younger vs older study participants (all $P > 0.05$).

Skin Displacement

The cranially directed skin displacement observed between baseline and Day 30 was 0.1 [0.9] mm (range, –1.92 to 2.81 mm) ($P = 0.381$) in the lateral face (ie, lateral to the line of ligaments) and 0.1 [0.9] mm (range, –2.18 to 2.30 mm) ($P = 0.410$) in the lower face (Supplemental Figures 1, 2), neither achieving statistical significance. No statistically significant differences in the measured skin displacement parameters were observed between younger vs older study participants (all $P > 0.05$).

DISCUSSION

This clinical interventional study was designed to investigate whether a novel neuromodulator injection algorithm had the ability to reposition the middle and lower facial soft tissues and to thus influence the aesthetic appearance 30 days after the treatment. The rationale behind this technique is the understanding of facial anatomy and its interconnected concepts of layered facial arrangement and tissue mobility.[12,17,22] It was theorized that by reducing the depressor effect of the platysma above the mandible, the novel injection algorithm may allow the elevator muscles of the face to predominate in effect, thereby increasing volume in the midface, repositioning the soft tissue of the lower face, and thus improving jawline contour. This injection algorithm did not aim to enhance the contour between the lower face and the neck and did not aim to smoothen the neck contour like the previously described Nefertiti lift injection technique or its subsequent iterations.[8-11] All effects observed in this study can be explained through the concepts of facial biomechanics. No adverse events were observed during the study period.

The objective 3D imaging analyses revealed a volume increase of the midface by 0.46 mL, ie, the facial soft tissues located cranial to a line connecting the oral commissure to the tragus and below a line connecting the

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lateral canthus and tragus (Figure 2). This increase in volume was additionally confirmed by the rating of the treating physician and an independent observer, although the semiquantitative scoring failed to reach statistical significance (change of 0.13 with $P = 0.096$). The clinical effects of the observed midfacial volume increase can be explained as follows: the neuromodulator treatment affected the contractility of the platysma which is the major facial depressor. The reduced caudal pull allowed an increased cranial pull of the facial elevators such as the orbicularis oculi, zygomaticus minor, and major muscles due to their connection via the SMAS to the platysma.[13-17] The resultant repositioning of the SMAS increased the tension of the transverse facial septum, allowing the septum to push the midfacial fat compartments more cranially. This cranial push of those fat compartments resulted in an increase in midfacial volume which was objectively measurable via 3D imaging. The fact that the semiquantitative scoring failed to reach statistical significance could potentially be due to the limited ability of raters to gauge volume increase in a specific facial region; it was, however, detected through quantitative imaging. Interestingly, the volume increase of 0.34 mL was predominantly observed medial to the line of ligaments,[21,23] which corresponds precisely to the location of the midfacial fat compartments. In the lateral face, however, no such volume increase was observed as here the soft tissue volume slightly reduced by –0.07 mL; this is in line with facial anatomic concepts because in the lateral face no deep facial fat compartments are present, only superficial fat compartments which lie above the SMAS.[24,25]

The results of this study provide clinical evidence that with neuromodulator injections alone, the lower and midfacial soft tissues can be repositioned toward a more cranial position. Objective 3D imaging data revealed that the volume lateral to the line of ligaments slightly decreased (–0.07 mL) and that skin displacement pointed in a cranial direction by 0.1 mm both in the middle and in the lower face. Although neither of the observed changes achieved statistical significance (when compared to no change as statistical reference), the results are trending towards cranial soft tissue repositioning. It could be argued that without statistical significance the differences are due to coincidence, but the direction of the skin displacement in the cranial direction is consistent with the volume decrease observed in the lower face and the volume increase in the midface. It should be added that a self-designed score for gummy-smile severity following the 5-point Likert scale design of all other scores used in this study increased significantly after the neuromodulator treatment. This increase in gingival show (as assessed during Duchenne-type smiling) is most likely due to a sudden imbalance between facial depressors (caudal pull of the platysma) and facial elevators favoring the elevators. This increase in cranial pull, effected by the zygomaticus minor, the levator labii superioris, and the SMAS, resulted in a statistically significant increase in gummy-smile severity which was observed in this study. Thus, injectors should exercise caution when performing this injection technique on patients with a gummy smile at baseline.

The repositioning effect of the performed injection algorithm was evaluated semiquantitatively via the jawline contouring scale, which revealed a statistically significant improvement of 0.78 on a 0 to 4 scale, best to worst. This was objectively confirmed by a decrease in lower facial volume of 0.30 mL for the entire lower face. Although the 3D imaging measurements did not reach statistical significance, the trend observed is in line with the previous measurements and consistent with the clinical observations: the decreased caudal pull of the platysma following the neuromodulator injection algorithm allows the facial elevators to predominate and reposition the lower and midfacial soft tissues more cranially, decreasing lower facial volume and increasing the midfacial volume especially in the medial face.

The 30-day follow-up period could be considered a limitation of this investigation because the typical duration of effect for neuromodulator treatments is usually 3 to 4 months. However, we sought to evaluate the potential clinical effects of this novel technique, not duration, and with the study design were able to see neuromodulator effect as soon as 14 days postinjection. A long follow-up period would likely show the reduction of the drug effect which would consequently result in the reduction of the soft tissue repositioning effects. We suspect that the effects seen during this study would be transient as is typical for facial aesthetic neuromodulator treatments. It is unclear which (if any) long-term benefits would result from the repeated treatments with the described injection technique; this could be the subject of future studies. Another limitation of the study is that at baseline a statistical difference was observed in the semiquantitative scores between patients treated with the 3 types of toxins. This resulted in an inability to compare the different toxins. Because this was beyond the scope of this study, future investigations might be able to expand on the results presented herein and additionally compare different dosages. It could be speculated that a higher dose might result in a greater facial soft tissue repositioning effects. Another limitation of this study could be the selected age criterion for patient inclusion into this study (age, <50 years). It could be argued that at higher ages, muscle function is reduced and therefore the identified effect is only applicable to a younger study population. In the present study, n = 39 were older than the average age of the entire population (≥38 years of age) and stratified statistical analyses revealed that when

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comparing the change of each evaluated score (clinical and 3D), older individuals statistically significantly improved more (jawline contour) or their change was not inferior to younger study participants (≤37 years of age) in all other assessed scores. In no evaluated score did older study participants display a statistically significant worse improvement than younger study participants. This fact could be supported by a recent investigation utilizing surface-derived noninvasive electromyography which revealed that facial muscles do not show changes at higher ages.[26] Another limitation of the study could be that the assessment was performed by the treating physician themselves and an independent observer, the latter being a colleague or an assistant to the injector. It is commonly assumed that an assessment carried out by the treating physician can be influenced toward a more positive outcome. It could be debated, however, that minute changes can best be observed by the treating physician only, whereas those small changes could be overlooked by an independent or by a nontreating observer. A bias toward a better outcome when evaluated by the injector is most applicable for fillers or for other immediately effective aesthetic treatments. The neuromodulator treatment algorithm presented in this study was assessed at Day 14 and Day 30. It could be argued that the treating physician had no recollection of the score at baseline or of the appearance of the patient at baseline, especially because the rating was not performed pairwise, therefore allowing for increased objectivity during the posttreatment evaluation.

In contrast to previous publications,[27,28] the presented neuromodulator injection algorithm resulted in no adverse events during the course of the study. This can be attributed to the superficial product administration and to the laterally oriented injection direction. No dysphagia was observed because the injection sites are all located 1 cm cranial to the superior border of the bony mandibular rim. This prevents product diffusion toward the suprahyoid or infrahyoid muscles which can result in difficulties during swallowing.

CONCLUSIONS

This clinical interventional prospective study sought to evaluate the effects of a novel neuromodulator injection algorithm. The results revealed that the facial soft tissues were repositioned during the 30-day follow-up period following the treatment; this was reflected through an increase in midfacial volume as well as through a decrease in lower facial volume when measured via 3D-scan-based volumetric analyses. Additionally, on live clinical assessment an increase in midfacial volume and an increase in jawline contouring was observed. No adverse events were reported over the course of the study. According to these data, the presented neuromodulator injection algorithm is a safe and effective option for patients desiring greater midfacial fullness and an enhancement of jawline contour without injectable fillers or surgical intervention.

Supplemental Material

This article contains supplemental material located online at www.aestheticsurgeryjournal.com.

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